

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from volume and load it into delta tables for customer and order data.

Preconditions:

CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a delta table named "customer".
2. Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a delta table named "order".
3. Add a new column "TotalAmount" to the "order" delta table, which is the product of "PricePerUnit" and "Qty" columns.

FR-DLT-002: Cleanse Customer and Order Data

Title: Cleanse Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order delta tables.

Preconditions:

- Customer and order delta tables are loaded with data

Main Flow / Functional Steps:

1. Remove rows with null values from the "customer" and "order" delta tables.
2. Remove duplicate rows from the "customer" and "order" delta tables.

FR-DLT-003: Create ordersummary Table and Load Data

Title: Create ordersummary Table and Load Data

Description: The system must create an "ordersummary" table if it does not exist and load data into it by joining customer and order delta tables.

Preconditions:

- Customer and order delta tables are cleansed
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "ordersummary" table in the catalog "gen_ai_poc_databrickscoe" and schema "sdhc_wizard" if it does not exist.
2. Join the "customer" and "order" delta tables on the "CustId" column.
3. Load the joined data into the "ordersummary" table, which is an SCD type 2 table.

FR-DLT-004: Update ordersummary Table on Customer Data Change

Title: Update ordersummary Table on Customer Data Change

Description: The system must update the "ordersummary" SCD type 2 table when there is a change in the customer table.

Preconditions:

- "ordersummary" table is loaded with data
- Customer data is updated

Main Flow / Functional Steps:

1. Identify changes in the "customer" table.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update the "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-005: Create customeraggregatespend Table and Load Data

Title: Create customeraggregatespend Table and Load Data

Description: The system must create a "customeraggregatespend" table and load aggregated data from the "ordersummary" table.

Preconditions:

- "ordersummary" table is loaded with data

- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
2. Aggregate the "TotalAmount" column from the "ordersummary" table by grouping by "Name" and "Date" columns.
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement the above steps using Delta Live Tables.

Preconditions:

- All previous steps are completed

Main Flow / Functional Steps:

1. Implement the above steps using Delta Live Tables to automate the data pipeline.